

Bisection Method

A simple numerical method for finding a root of $f(x) = 0$

Main idea: keep cutting the interval in half



Used when exact algebraic solution is difficult.

Works by finding two points a and b where $f(a)$ and $f(b)$ have different signs.

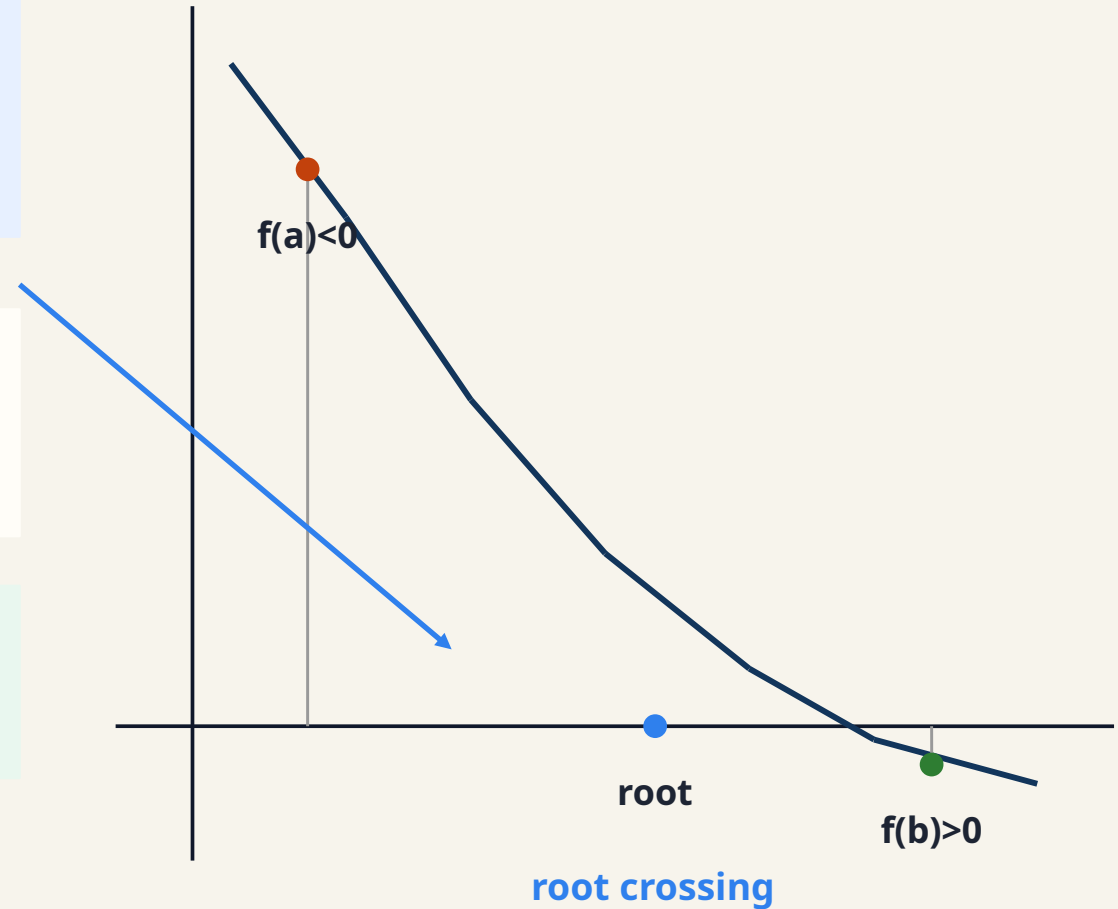
Then the root must lie somewhere between a and b .

Key idea: opposite signs mean a root is inside

If $f(x)$ is continuous between a and b and $f(a) \times f(b) < 0$, then at least one root exists between a and b .

Why? A continuous curve cannot jump from negative to positive without crossing the x-axis.

Crossing the x-axis means $f(x) = 0$, so that x -value is a root.



Bisection Method in 4 simple steps

1
Choose a and b
with opposite
signs

2
Find midpoint
 $x_r = (a+b)/2$

3
Check the sign
of $f(x_r)$

4
Keep the half
that still has the root

Example interval after one bisection



If $f(a) \times f(x_r) < 0$, root is in $[a, x_r]$

Otherwise, root is in $[x_r, b]$

How to choose the next interval

After calculating x_r , compare the signs.

Case A
 $f(a) \times f(x_r) < 0$

Root is between a and x_r
New interval: $[a, x_r]$
Set $b = x_r$

Case B
 $f(a) \times f(x_r) > 0$

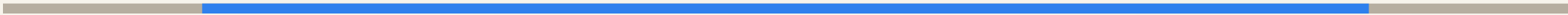
Root is between x_r and b
New interval: $[x_r, b]$
Set $a = x_r$

Case C
 $f(a) \times f(x_r) = 0$

x_r is exactly the root
Stop the calculation

The interval becomes smaller every iteration

whole interval



half



smaller half



How many iterations are needed?

The interval length becomes half each time.

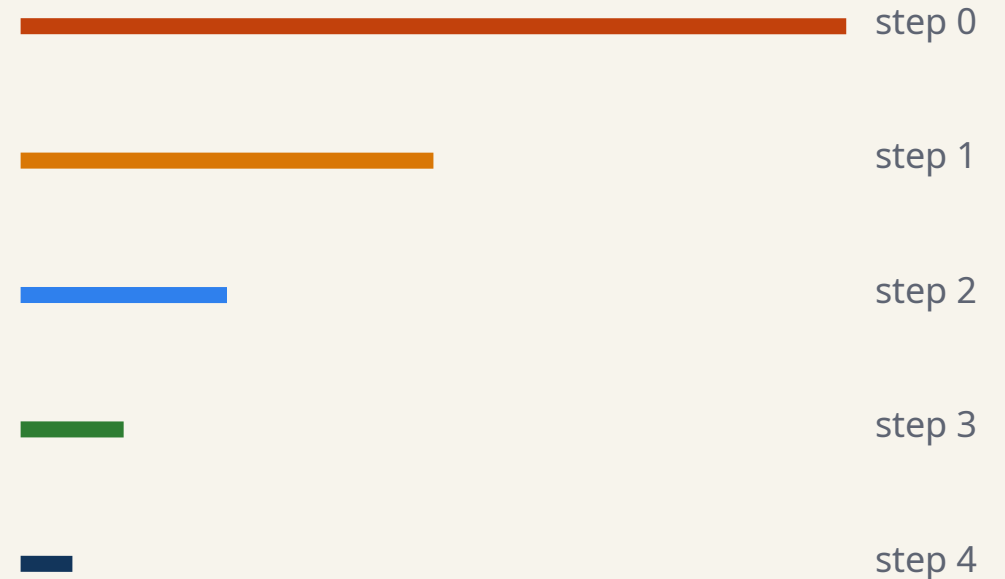
After n steps, interval length = $|b - a| / 2^n$

To achieve accuracy ϵ , make the interval small enough:

$$|b - a| / 2^n \leq \epsilon$$

$$n \geq \log_{\epsilon}(|b - a| / \epsilon) / \log_{\epsilon} 2$$

Halving effect



smaller interval = better root estimate

Example from the book

Given: $|b - a| = 1$ and $\varepsilon = 0.001$

Use:

$$n \geq \log(1 / 0.001) / \log(2)$$

$$n \geq \log(1000) / \log(2) \approx 9.97$$

Therefore, choose $n \geq 10$
iterations

Meaning: 10 halvings make the interval
very small



When should we stop?

In real problems, the exact root may not appear. So we stop when the answer is “accurate enough.”

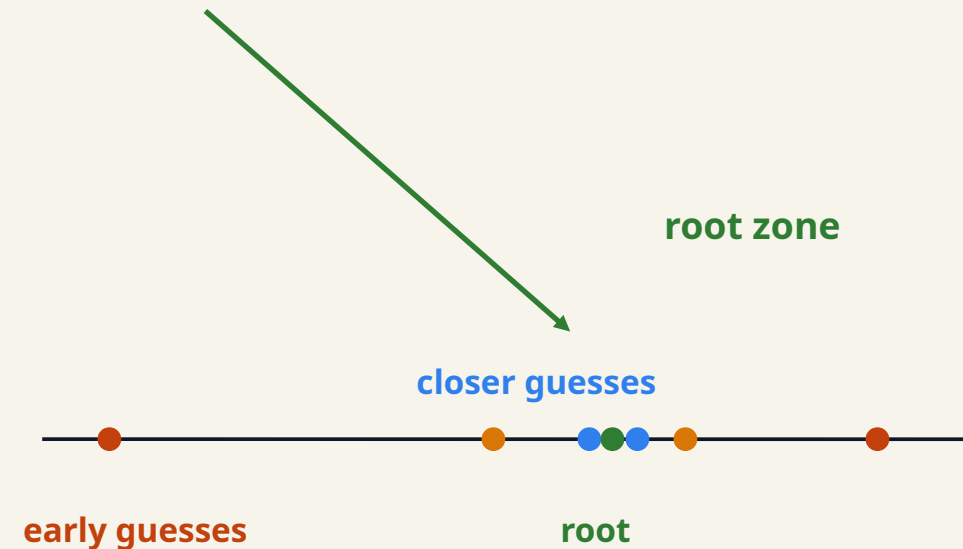
Approximate percentage error:

$$e_r = |(x_r' - x_r) / x_r'| \times 100\%$$

Stop when e_r is less than the chosen tolerance e_p .

Successive estimates get closer

The gap between new and previous estimates becomes smaller.



Bisection Method: quick summary

1. Start with opposite signs
 $f(a) \times f(b) < 0$

2. Midpoint is the next guess
 $x_r = (a+b)/2$

3. Repeat until accurate enough interval or error is small

Advantages

- Very simple
- Always converges if the sign condition holds
- Easy to program

Limitations

- Can be slow
- Needs a bracket $[a,b]$
- Finds one root inside the interval

Final concept: Keep only the half of the interval where the root still exists.